25901236 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Skill Enhancement Course (SEC) - MG4SECECT201 - PCB DESIGN AND 3D PRINTING

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

MCQ

Answer all questions

Each question carries **1** marks

1. In Through-Hole Technology (THT), component leads are: [U] / [CO1]

   a). Placed on PCB surface b). Inserted into drilled holes

   c). Glued to PCB d). Press-fitted

2. Which of the following is an electrical design rule? [U] / [CO1]

   a). Silkscreen font size b). Trace width and spacing

   c). Board color d). Component labeling

3. Layers in multilayer PCB can go up to: [U] / [CO1]

   a). 10 b). 20

   c). 30 d). 40

4. The unit of inductance is: [U] / [CO1]

a). Farad b). Henry

c). Ohm d). Volt

5. Altium supports 3D view of PCB: [U] / [CO1]

a). Yes b). No

c). Sometimes d). Only externally

6. Proteus is mostly used before: [U] / [CO1]

a). Coding b). Manufacturing

c). Selling d). Marketing

7. In a schematic, wires are also called: [U] / [CO1]

a). Pads b). Vias

c). Nets d). Traces

8. EAGLE stands for: [U] / [CO1]

a). Easy Application for Graphic Layout Editor

b). Electronic Automation Graphic Layout Editor

c). Electrical Analog Graphic Layout Editor

d). Electronic Application Guide Layout Editor

9. Which component is commonly used for manual ON/OFF control in a circuit? [U] / [CO1]

a). Sensor b). Transformer

c). Switch d). Integrated Circuit

10. Who is credited with developing the first Printed Circuit Board? [U] / [CO1]

a). Thomas Edison b). Nikola Tesla

c). Paul Eisler d). Alexander Graham Bell

11. Which part of the PCB provides mechanical support? [U] / [CO1]

a). Copper layer b). Silkscreen

c). Substrate d). Solder mask

12. Compared to single sided PCB, double sided PCB is: [U] / [CO1]

a). Simpler b). More complex

c). Cheaper d). Less flexible

13. Current sense resistors usually have resistance values: [U] / [CO1]

a). Above 1 kΩ b). Less than 1 ohm

c). In megaohms d). In gigaohms

14. Vias are mainly used in: [U] / [CO1]

a). Single-sided PCB b). Double-sided PCB

c). Multilayer PCB d). Both B and C

15. In a single sided PCB, components are placed on: [U] / [CO1]

a). One side only b). Inside the board

c). Copper side d). Both sides

16. EDA software helps in drawing: [U] / [CO1]

a). Circuit diagrams b). Flowcharts

c). Building plans d). Graphs

17. After schematic capture, the next main step is: [U] / [CO1]

a). Fabrication b). Simulation

c). Validation d). Board layout

18. PCB prototypes are usually made to: [U] / [CO1]

a). Increase production cost b). Replace schematic capture

c). Change PCB color d). Test and verify the design before mass production

19. Gerber files are used for: [U] / [CO1]

a). Simulation b). Circuit testing

c). PCB fabrication d). Component placement

20. Electronic components are mainly classified into: [U] / [CO1]

a). Digital and analog b). Active and passive

c). Input and output d). AC and DC

21. One disadvantage of FDM printing is: [U] / [CO2]

a). High cost b). Smooth surface finish

c). Visible layer lines d). Limited build size

22. Infill density mainly affects the: [U] / [CO2]

a). Color of the print b). File format

c). Printer speed d). Strength and weight of the object

23. TinkerCad is mainly accessed through: [U] / [CO2]

a). Offline desktop app only b). Mobile application

c). Web browser d). Command line

24. Which CAD software is beginner-friendly and web-based? [U] / [CO2]

a). SolidWorks b). Blender

c). Fusion 360 d). TinkerCAD

25. The purpose of a gear mechanism project is to test: [U] / [CO2]

a). Motion and alignment b). Color quality

c). Sound output d). File size

26. Which device is required to use CAD software? [U] / [CO2]

a). Printer b). Scanner

c). Computer d). Projector

27. Selective Laser Sintering (SLS) works by: [U] / [CO2]

a). Melting filament b). Curing resin

c). Fusing powder with laser d). Spraying material

28. For SLA printing, the minimum wall thickness is typically: [U] / [CO2]

a). 0.5–1 mm b). 0.1 mm

c). 2–3 mm d). 1–3 mm

29. Which printing method is the most expensive among FDM, SLA, and SLS? [U] / [CO2]

a). FDM b). SLA

c). SLS d). All are equal

30. Designing for functionality includes considering: [U] / [CO2]

a). Tolerances and fit b). Color themes

c). File names d). Printer brand

31. Fusion 360 can be used for: [U] / [CO2]

a). 3D modeling only b). Web design

c). CAD, CAM, and CAE d). Mobile app development

32. Which software is web-based and easy to learn? [U] / [CO2]

a). TinkerCad b). Fusion 360

c). AutoCAD d). SolidWorks

33. Overhang angles greater than ____ usually require supports. [U] / [CO2]

a). 15° b). 30°

c). 45° d). 90°

34. In FDM printing, material is deposited: [U] / [CO2]

a). As a single block b). Randomly

c). In liquid form d). Layer by layer

35. What is 3D printing also known as? [U] / [CO2]

a). Additive manufacturing b). Subtractive manufacturing

c). Injection molding d). Casting

[1x35 = 35]

Part B

Short Essay Questions

Answer any **3** questions

Each question carries **5** marks

36. Describe the key technical challenges involved in system integration projects. [An] / [CO3]

37. Identify operational challenges during integration implementation and provide solutions. [An] / [CO3]

38. Discuss the major stakeholder management challenges in integration initiatives and suggest suitable solutions. [An] / [CO3]

39. Describe in detail a Smart City integration case study and explain its major components and outcomes. [An] / [CO3]

40. Analyze a real-world example of integration in the education sector and its impact. [An] / [CO3]

[5x3 = 15]